

Name : Ragini Singh

Cohort : JH

Roll No : 150096724023

Sub : Chemistry Assignment 12

Chemistry Assignment - 12
Name - Ragini Singh Roll No - 023 Date - 4.5.20

1. Rate = $k[RX]$
Rate = k [S. Substrate]
Follows first-order kinetics
The rate depends only on the concentration of the substrate (ALKYL HALIDE).

2. Such reactions take place in two steps.
 $R-X \xrightarrow{\text{slow}} R^+ + X^-; R^+ + Y^- \xrightarrow{\text{fast}} R-Y$
 $3^\circ > 2^\circ > 1^\circ$ and CH_3X do not go.
• SN_1 mechanism (two step, stepwise).
• The leaving group departs first, forming a flat, planar, carbocation intermediate. This slow step determines the reaction rate.
• A nucleophile rapidly attacks the carbocation from either side (front or back), leading to a racemic mixture.
3. Strength of nucleophile not important; good ionising solvent required.
4. possible rearrangements lead to racemisation.

Rate = $k[RX][Nu^-]$
Rate = k [S. Substrate] [S. Nucleophile]
Follows second-order kinetics.
The rate depends on the concentration of both the substrate and the NUCLEOPHILE.
Such reactions take place in SINGLE steps as bond forming & breaking occurs simultaneously.
 $Y^- + R-X \xrightarrow{\text{slow}} [Y \cdots R \cdots X]^\ddagger \xrightarrow{\text{fast}} Y-R + X^-$
 $CH_3X > 1^\circ > 2^\circ > 3^\circ$
• The NUCLEOPHILE attacks from the back side.
• while the leaving group simultaneously departs.
• it passes through an unstable, single transition state, resulting in a complete inversion of configuration (Walden inversion).
- strong nucleophiles required
- may go faster in less polar solvent
- No rearrangements
- lead to inversion.

1S - example
 $CH_3CH_2CH_2Br \xrightarrow{\text{slow}} CH_3CH_2CH_2^+ + Br^-$
 $CH_3CH_2CH_2^+ + OH^- \xrightarrow{\text{fast}} CH_3CH_2CH_2OH$
- 2S ex: $OH^- + CH_3CH_2CH_2Br \xrightarrow{\text{fast}} CH_3CH_2CH_2OH + Br^-$
tert. butyl bromide fast. SN1 carbocation